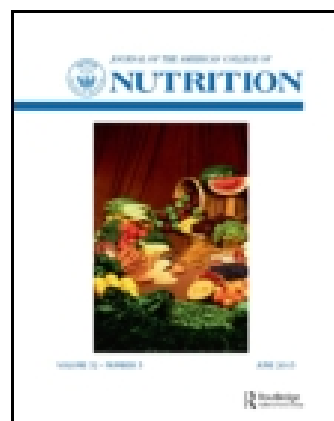


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Thiamin, Riboflavin and Vitamin B₆: Impact of Restricted Intake on Physical Performance in Man

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Key words: thiamin, riboflavin, vitamin B₆, vitamin restriction, physical performance

Objective: A combined marginally deficient status of thiamin, riboflavin, vitamin B₆ and vitamin C may affect physical performance, but the relative contribution of each vitamin can only be speculated. In a previous study we did not find any effect of restricted intake of vitamin C individually. Therefore, the functional effect of restriction of thiamin, riboflavin or vitamin B₆, individually or in conjunction, was investigated.

Methods: A double-blind, 2 × 2 × 2 complete factorial experiment on the effects of thiamin, riboflavin and vitamin B₆ restriction on physical performance was executed with 24 healthy men. During 11 weeks of low vitamin intake, the subjects were given a daily diet of regular food products providing no more than 55% of the Dutch Recommended Dietary Allowances (RDA) for thiamin, riboflavin and vitamin B₆. Other vitamins were supplemented at twice the RDA level.

Results: In vitamin-restricted subjects, blood vitamin levels, erythrocytic enzyme activities and urinary vitamin excretion decreased and in vitro erythrocytic enzyme stimulation increased. Short-time vitamin restriction had no harmful effects on health. A significant overall decrease was observed in aerobic power ($\dot{V}O_2$ -max; 11.6%), onset of blood lactate accumulation (OBLA; 7.0%) and oxygen consumption at this power output ($\dot{V}O_2$ -OBLA; 12.0%), peak power (9.3%), mean power (6.9%) and related variables ($p < 0.01$). However, the observed performance decrements could not be attributed to marginal deficiency for any of the vitamins studied.

Conclusion: The absence of vitamin-specific effects on performance decrements due to thiamin, riboflavin and vitamin B₆ restriction suggests quantitatively similar but non-additive effects of these B-vitamins on mitochondrial metabolism.

INTRODUCTION

It is well recognized that severe deficiency of vitamins impairs various human functions—physiological functions such as exercise performance, immunocompetence, cognitive function and psychomotor performance as well as mood and personality [1]. However, knowledge of the functional significance of marginal vitamin deficiency is still limited. Marginal deficiency is characterized by vitamin levels in body fluids and tissues below reference limits and by biochemical changes in metabolism without any vitamin-specific clinical signs and symptoms of deficiency [2]. We observed in a randomized, controlled depletion experiment with young volunteers that a combined mar-

ginally deficient status of thiamin, riboflavin, vitamin B₆ and vitamin C was associated with a decrease of aerobic power and onset of blood lactate accumulation (OBLA), without cognitive functions being affected [3]. These vitamins are involved as cofactors in mitochondrial metabolism [4,5]. However, little is known about the relative contribution of each individual vitamin to the detrimental effect of the combined marginal deficiency. Recently, we observed that a marginally deficient vitamin C status, apart from an increased heart rate at OBLA level, does not affect physical performance [6]. Our observations suggest either an effect of any of the B-vitamins or an interaction between two or more vitamins. Therefore, in the present study a factorial experiment was designed to investigate whether

Abbreviations: EGOT-erythrocyte glutamic-oxaloacetic transaminase, EGR-erythrocyte glutathion reductase, ETK-erythrocyte transketolase, FAD-flavin adenine dinucleotide, OBLA-onset of blood lactate accumulation, PLP-pyridoxal 5'-phosphate, RER-respiratory gas exchange ratio, RPE-rate of perceived exertion, ThDP-thiamin diphosphate, VE-ventilatory equivalent

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restriction of thiamin, riboflavin or vitamin B₆, individually or in conjunction, is associated with a decrease of physical performance.

A factorial approach was chosen because factorial experiments require fewer experimental subjects to yield equal precision as single-factor approaches, allow for the investigation of interactions among factors, and lead to results that apply over a wide range of conditions [7].

We examined the decrease in thiamin, riboflavin and vitamin B₆ status to marginal or deficient levels resulted from a diet composed of normal food products which provided 26–59% of the Dutch Recommended Dietary Allowances (RDA) [8] for the B-vitamins in question.

METHODS

Subjects and Experimental Design

The study consisted of a 14-week, double-blind, 2 × 2 × 2 complete factorial experiment [7]. The factorial experiment is a commonly used statistical design which enables a researcher to identify the main effects of a number of factors (deficiency in thiamin, riboflavin and vitamin B₆) as well as the interactive effects of these factors (e.g., when the effect of thiamin deficiency is dependent on riboflavin deficiency). The factorial experiment is, under assumption of additivity of effects, the most efficient design for detecting main effects and interactive effects of several factors.

Twenty-four subjects were matched for aerobic power as measured a few weeks before the start of the experiment and were randomly assigned to the eight treatment groups, consisting of all combinations of deficiency/non-deficiency in each of the vitamins (Table 1). Each of these 8 groups contained 3 subjects. In the analysis of the factorial experiment with ANOVA, the data of all 24 subjects are used for the tests of all main effects and interactions. In the comparison of the effect of one vitamin, the two groups that are compared are identical with respect to deficiency in the other vitamins due to the factorial design.

An 11-week period of low vitamin intake (experimental or restriction period) was preceded by a 1-week introductory period and followed by a 2-week concluding period. The 24 subjects were apparently healthy men (aged 23.4 ± 2.6; body weight 74.7 ± 7.7 kg; $\dot{V}O_2$ -max 3.76 ± 0.44 l/min). The subjects had normal Dutch dietary habits and showed moderate drinking and smoking behavior. All subjects were moderately active and were instructed not to change their daily activity pattern during the study.

During the study all subjects remained in their own environment. Dinners had to be consumed in the metabolic ward of the Institute each day; other meals were consumed at home.

The experiment was carried out in conformity with the rules of the Declaration of Helsinki and approved by the

Table 1. Factorial Design (n = 24)

Subject	Thiamin	Riboflavin	Viamin B ₆
1	–	–	+
2	–	+	–
3	+	+	–
4	+	+	+
5	+	–	–
6	–	+	–
7	+	+	–
8	–	–	–
9	+	–	+
10	+	+	+
11	–	+	+
12	–	–	+
13	+	–	+
14	+	+	+
15	–	+	–
16	+	–	–
17	–	+	+
18	+	–	–
19	–	–	–
20	+	+	–
21	–	–	–
22	+	–	+
23	–	+	+
24	–	–	+

+ = restricted
– = not-restricted

Institute’s external medical ethical committee. Informed written consent was obtained from each volunteer.

Dietary Methodology

To get accustomed to the experimental diet all subjects received the diet in the introductory period (week-1). The experimental diet was composed of common food products and was consumed during the remaining part of the study. The composition of this diet is presented in Table 2. The dietary characteristics, based on chemical analysis of three daily samples, are given in Table 3. The energy content of the experimental diet was adjusted to the mean energy requirement of the study population derived from a dietary history before the study. During the introductory period (week-1) all subjects were supplemented with twice the Dutch Recommended Dietary Allowances (RDA) for all vitamins except vitamin K. This vitamin dosage was chosen ensure that no other vitamin deficiencies could confound the results. This regimen was continued in the unrestricted subgroup until the end of the experiment. In the experimental period (weeks 0–11), the 7 restricted subgroups received the same supplement except for one or more of the B-vitamins (Table 4). In the concluding period (weeks 12 and 13), all subjects received the same supplement as in the introductory period. Vitamin supplements were consumed during dinner under supervision to ensure

Table 2. Composition (g or mL) of the Experimental Diet (mean daily values)

Breakfast	
two white-bread rolls	90
dietary margarine	
cheese	20
jam	14
Lunch	
four white-bread rolls	180
dietary margarine	
cheese	20
smoked beef	15
jam	14
peanut butter	10
Dinner	
white rice	200
carrots	100 on days 1, 3, 5, etc
green peas	50
French beans	150 on days 2, 4, 6, etc
dietary margarine	10
apple sauce	100
beef	100
white-cream or curry sauce	100
whipped cream	15
canned peaches or pears	100
Allowances during the day	
dietary margarine	50
tea (instant), 3 cups	0.9
coffee (instant), 4 cups	6.0
sugar	28
powdered milk	12
soft drinks	500
Dutch honey cake	40
cookies	20

that each subject received the correct supplement. Mineral and trace element supplementations were equal for all subjects throughout the study, i.e., 100% of the RDA, as described previously [3].

Table 3. Experimental Diet Characteristics (mean daily values as analyzed)

Energy	
(MJ/day)	11.2
(kcal/day)	2679
Total fat (g/day)	102.1 (34.3 en%)
Linoleic acid (en%)*	11.5
Saturated fatty acids (en%)	13.2
Total available carbohydrates (g/day)	372 (55.5 en%)
Total protein (g/day)	68.1 (10.2 en%)
Thiamin (mg/day)	0.43 (39.1% of RDA)
Riboflavin (mg/day)	0.94 (58.8% of RDA)
Vitamin B ₆ (mg/day)	0.39 (26.0% of RDA)
Vitamin C (mg/day)	79.4 (158.8% of RDA)

*en%, % of total energy intake.

Table 4. Composition of the Vitamin Supplements*

Vitamin	Capsule	No.	1	2	3	4	5	6	7	8
All-trans-retinyl palmitate	5660	IU	+	+	+	+	+	+	+	+
Cholecalciferol	400	IU	+	+	+	+	+	+	+	+
d,l- α -tocopheryl acetate	30	mg	+	+	+	+	+	+	+	+
Cyanocobalamin	6	μ g	+	+	+	+	+	+	+	+
Folic acid	800	μ g	+	+	+	+	+	+	+	+
L-ascorbic acid	100	mg	+	+	+	+	+	+	+	+
Nicotinamide	42	mg	+	+	+	+	+	+	+	+
Calcium pantothenate	20	mg	+	+	+	+	+	+	+	+
Thiamin mononitrate	2.5	mg	+	-	+	+	-	-	+	-
Riboflavin	4	mg	+	+	-	+	-	+	-	-
Pyridoxine hydrochloride	4	mg	+	+	+	-	+	-	-	-

Lactose was added to make the final weight approximately 360 mg.

*+ = present; - = not present.

Blood and Urine Analyses

Blood was withdrawn from an antecubital vein after an overnight fast at the start of the experimental period (week 0) and in weeks 5 and 11. Whole blood samples were used to determine vitamin status. Thiamin status was assessed by means of total thiamin and erythrocyte transketolase (ETK) activity and in vitro ETK stimulation by thiamin diphosphate (ThDP) (α -ETK or activation coefficient) [9,10]. Riboflavin status was assessed by means of flavin adenine dinucleotide (FAD) and erythrocyte glutathione reductase (EGR) activity and in vitro EGR stimulation by FAD (α -EGR) [11,12]. Vitamin B₆ status was assessed by means of pyridoxal 5'-phosphate (PLP) and erythrocyte glutamate oxaloacetate (EGOT) activity and in vitro EGOT stimulation by PLP (α -EGOT) [13,14]. Vitamin C status was assessed by measuring the total vitamin C concentration, as an indicator vitamin, in whole blood [15]. On the day preceding blood sampling, 24-hour urine was collected for analyzing thiamin, riboflavin and 4-pyridoxic acid excretion [16,17]. Iron status was evaluated by determining the hemoglobin (Hb) concentration, the hematocrit (Ht) reading, and the number of erythrocytes in whole blood by using a semi-automatic hematology analyzer (Sysmex CC-180, Toa Medical Electronics, Kobe, Japan). Serum ferritin was assessed by an IRMA (immunoradiometric) method [18] with commercial reagents (Ramco Laboratories, Houston, TX).

Subject Health and Activity Sheet

Each subject completed a daily health and activity sheet, pertaining to physical activities, health, well-being, stress and hours of sleep, and recorded his weight on the sheet.

Physical Performance

Physical performance, assessed at the start of the experimental period (week 0) as well in weeks 5 and 11, was

quantified by measuring aerobic power ($\dot{V}O_{2\text{-max}}$) [19]; onset of blood lactate accumulation (OBLA) [20,21], defined as the power output corresponding to a level of 4 mmol lactate per liter whole blood or plasma; and anaerobic power. The anaerobic power tests were executed 2 days before the determination of OBLA and aerobic power. Each subject was familiarized with the tests in week 1.

Aerobic Power and Onset of Blood Lactate Accumulation (OBLA)

Aerobic power and OBLA were determined by incremental exercise on an electrically braked bicycle ergometer (Lode-HL-600R; Lode, Groningen, Netherlands). The subjects started cycling at 50 W and pedaled at a rate of 60–80 revolutions/minute (rpm). The workload was increased by 50 W after 2 minutes, and thereafter 25 W increments were given every 2 minutes until exhaustion. At the end of each step, the rating of perceived exertion (RPE) was measured on a 15-point scale [22]. During the test, ventilatory and gas exchange parameters were monitored every 30 seconds by an open-circuit sampling system (Oxycon-4, Mijnhardt, Bunnik, Netherlands). The gas analyzers were calibrated against standardized gases prior to testing; the dry gas meter was calibrated weekly with a 3-liter syringe (Pul Mark 1, RNA Medical Corp, Woburn, MA). Subjects breathed through a low-resistance breathing valve (# 2700, Hans Rudolph, Kansas City, MO). Heart rate was continuously monitored by an ECG with CM-5 lead positions. If desired, facial cooling was applied by means of a fan. As cooling may affect heart frequency [23], each subject had to choose during the first exercise test whether or not to use the ventilator during the study. During the test blood was sampled from a heparinized (12.5 U/5 mL), 18-gauge indwelling catheter (Quick-Cath, Travenol Laboratories, Castlebar, Ireland) with a three-way stopcock (Robinet 3 Voies, Vermed, Angers, France) which was inserted into an antecubital vein. Blood samples were drawn during the last 30 seconds of each exercise step. Blood lactate was determined in plasma by an enzymatic-calorimetric autoanalyzer method with commercial reagents (Boehringer Mannheim GmbH Diagnostica, # 256773, FRG). The exercise effort was considered maximal if the following criteria were fulfilled: a respiratory equivalent ($\dot{V}E/\dot{V}O_2$) above 30 [24]; a respiratory exchange ratio (RER) > 1.0 [25], and a heart rate within the 95% confidence interval of the age-related heart rate at maximum workload [26].

Anaerobic Power

Anaerobic power was assessed by means of the Wingate test [27], a 30-second supramaximal cycle ergometer test. This test was performed on a mechanically braked bicycle ergometer (Monark-Crescent, Varburg, Sweden) that had

been modified with a lever arm to which weights were attached for instantaneous application of resistance [28]. Revolutions were counted by means of an electronic device on the flywheel connected to a counter which registered the instantaneous, peak, and successive 5-second mean frequencies. Prior to the test, subjects warmed up for 2–4 minutes on the Lode electrically braked bicycle ergometer. They were then seated on the modified ergometer with the feet fastened in toe clips and straps; the seat height was adjusted such that there was a slight bend at the knee upon extension. The subjects were instructed to pedal at an initial rate of 100–120 rpm without resistance, enabling them to overcome the inertial resistance of the flywheel. On the command “ready, go”, they started to pedal as fast as possible and the lever arm was lowered, applying the load, and simultaneously the counter started timing the test and counting flywheel revolutions. The resistance applied was 0.075 kg/kg body weight for all subjects. At the end of the 30-second ergometer test the following variables were calculated: peak power, the highest power output during any 5-second period; mean power, the average power output during the entire 30 seconds; and power decrease, the difference between peak power and the lowest 5-second power output divided by the elapsed time. All three indices were also expressed in relative terms by dividing them by body weight. Peak power was assumed to reflect the ability of limb muscles to produce mechanical power in a short time. Mean power was assumed to reflect the ability of limb musculature to sustain extremely high power. It is assumed that power decrease reflects the rate of fatigue [27].

Statistical Methods

The method used consisted of an analysis of variance (ANOVA) of coefficients of fitted orthogonal polynomials over the experimental period (weeks 0 through 11) for each individual [29,30]. Independent estimates of the coefficients for the average level of a variable during the experimental period, and for linear and quadratic contrasts, were analyzed according to the factorial design of the study using ANOVA. In the ANOVA, the main effects for the factors thiamin, riboflavin and vitamin B₆ were tested, as well as the three 2-factor interactions and the 3-factor interaction. The linear coefficients indicate the linear changes of a variable over the period of low vitamin intake in units per week. Positive quadratic coefficients indicate a downward curvature, while negative quadratic coefficients indicate an upward curvature. In addition to the differences between coefficients, it was investigated whether the (overall) average coefficients differed significantly from zero by means of t-tests applying the pooled standard errors of the coefficients. Effects were noticed when the F value from the ANOVA exceeded the critical

value corresponding to a significance level of 1% to prevent type I errors due to the large number of variables tested. The levels of the results (mean $\pm$ SEM) in the experimental period (weeks 0, 5 and 11) are presented, as well as the coefficients of linear change in time (slopes) for the specific vitamin factors that affected a variable significantly. Where no main effects were found, the pooled mean levels and overall coefficients of linear change are presented.

RESULTS

Mean values and SEMs of both the restriction and control groups at the start (week 0), after 5 weeks (week 5), and at the end of the experimental period (week 11) are given in Tables 5, 6, 7, 8 and 9. In addition, the slopes (average change per week) and their SEMs for the restriction and the control groups are presented. Where no main effects were observed the pooled results are presented in the same way. The non-restricted (as to a particular vitamin), adequately nourished groups will be referred to as control groups consistently. In other words, controls for one vitamin may have had restrictions for one or more other vitamins. This is a characteristic feature of the factorial design.

In the experimental period, no significant vitamin-specific main effects or interactions in body composition were observed. However, a pooled linear decrease in body weight, percentage of body fat and fat-free mass was ob-

served ($p < 0.01$). The mean decrease in body weight was 5.4% (range -12.4% to -0.1%). The mean decreases in percentage of body fat and fat-free mass were 11.2% (range $+11.9\%$ to -43.8%) and 3.2% (range $+1.2\%$ to -7.8%), respectively.

Iron status indicators did not show any vitamin-specific effects or interactions in the experimental period. However, pooled linear decreases in Hb concentration and Ht reading were observed ($p < 0.01$). The mean decreases in Hb concentration and Ht reading were 5.4% (range $+3.1\%$ to -12.5%) and 6.8% (range $+4.4\%$ to -17.0%), respectively. No changes in time were observed for ferritin and vitamin C.

Vitamin Variables in Blood

At the start of the experimental period five subjects in the riboflavin-restricted and six subjects in the unrestricted group showed deficient FAD levels. However, only five of them showed deficient FAD levels in the screening before the experiment. Although all biochemical variables changed during that period, the degree and speed of vitamin depletion in restricted individuals varied. All thiamin-restricted subjects reached deficient total thiamin concentrations and 83% reached deficient α -ETK levels in the period of low vitamin intake. Of the riboflavin-restricted subjects, 92% reached a deficient FAD level and none of them reached deficient α -EGR levels. None of the restricted subjects reached deficient PLP blood levels and

Table 5. Vitamin Levels in Blood According to Treatment: Mean Values (SEM) and Slopes (SEM) in the Restriction and Control Groups

Variable	Group*	n	Week of Study						Slope	
			0		5		11			
Thiamin (nmol/L)	B ₁ -restricted	12	124	(5.2)	87	(3.6)	76	(4.2)	-4.3191	(0.2969) [†] +
	B ₁ -control	12	124	(4.0)	134	(5.7)	132	(4.4)	0.6479	(0.3550)
ETK activity (U/mmol Hb)	B ₁ -restricted	12	13.0	(0.6)	9.0	(0.3)	8.0	(0.4)	-0.4463	(0.0384) [†] +
	B ₁ -control	12	13.1	(0.7)	12.6	(0.3)	12.8	(0.4)	-0.0307	(0.0565)
α-ETK	B ₁ -restricted	12	1.07	(0.02)	1.24	(0.02)	1.33	(0.04)	0.0236	(0.0031) [†] +
	B ₁ -control	12	1.08	(0.02)	1.08	(0.02)	1.09	(0.01)	0.0002	(0.0020)
FAD nmol/L	B ₂ -restricted	12	265	(16.6)	242	(8.3)	227	(7.3)	-3.4570	(1.3462) [†] +
	B ₂ -control	12	252	(10.4)	307	(12.3)	297	(9.6)	3.9011	(0.5408) [†]
EGR act. (U/mmol Hb)	B ₂ -restricted	12	139	(5.7)	90	(4.6)	84	(4.8)	-4.9002	(0.2083) [†] +
	B ₂ -control	12	146	(7.7)	149	(7.2)	156	(7.2)	0.8782	(0.1382) [†]
α-EGR	B ₂ -restricted	12	0.99	(0.01)	1.19	(0.02)	1.23	(0.01)	0.0210	(0.0013) [†] +
	B ₂ -control	12	0.99	(0.02)	0.97	(0.01)	0.96	(0.01)	-0.0030	(0.0011)
PLP (nmol/L)	B ₆ -restricted	12	94	(4.6)	45	(1.6)	38	(1.8)	-4.9574	(0.3561) [†] +
	B ₆ -control	12	93	(6.4)	112	(6.8)	112	(6.7)	1.6717	(0.3818) [†]
EGOT act. (U/mmol Hb)	B ₆ -restricted	12	70	(3.1)	56	(2.8)	46	(4.7)	-2.1864	(0.2825) [†] +
	B ₆ -control	12	75	(5.8)	74	(5.1)	82	(8.6)	0.6414	(0.3517)
α-EGOT	B ₆ -restricted	12	1.66	(0.04)	1.88	(0.05)	2.13	(0.11)	0.0430	(0.0085) [†] +
	B ₆ -control	12	1.63	(0.04)	1.63	(0.04)	1.63	(0.08)	0.0001	(0.0059)

* For brevity's sake, the obsolete designations B₁, B₂ are used for thiamin, riboflavin, respectively.

[†] Slope significantly different from zero ($p < 0.01$).

+ Significant difference in slopes between restriction and control groups ($p < 0.01$).

Table 6. Urinary Vitamin Excretion According to Treatment: Mean Values (SEM) and Slopes (SEM) in the Restriction and Control Groups

Variable	Group	n	Week of Study						slope	
			0		5		11			
Thiamin	B ₁ -restricted	12	327	(42.3)	21	(3.9)	11	(3.2)	-27.7967	(3.7257) ^{†*}
(μ/24 hours)	B ₁ -control	12	325	(42.0)	424	(52.1)	332	(38.6)	0.1177	(5.0725)
Thiamin	B ₁ -restricted		0.9703	(0.1255)	0.0623	(0.0116)	0.0326	(0.0096)	-0.0825	(0.0111) ^{†*}
(μmol/24 h)	B ₁ -control		0.9644	(0.1246)	1.2582	(0.1546)	0.9825	(0.1145)	0.0003	(0.0151)
Riboflavin	B ₂ -restricted	12	2701	(188.1)	22	(4.0)	11	(2.2)	-236.5426	(16.451) ^{†*}
(μg/24 h)	B ₂ -control	12	2336	(390.8)	1708	(208.1)	1744	(239.2)	-51.3118	(42.1609)
Riboflavin	B ₂ -restricted		7.1835	(0.5003)	0.5858	(0.0106)	0.0293	(0.0059)	-0.6291	(0.0438) ^{†*}
(μmol/24 h)	B ₂ -control		6.2128	(1.0394)	4.5426	(0.5535)	4.6383	(0.6362)	-0.1365	(0.1121)
4-Pyridoxic	B ₆ -restricted	12	2565	(162.0)	302	(17.8)	250	(16.3)	-203.8571	(14.0810) ^{†*}
acid (μg/24 h)	B ₆ -control	12	2408	(108.4)	2367	(199.5)	3042	(165.2)	59.4803	(21.1744)
4-Pyridoxic	B ₆ -restricted		14.0164	(0.8852)	1.6503	(0.0973)	1.3661	(0.0891)	-1.1140	(0.0769) ^{†*}
acid (μmol/24 h)	B ₆ -control		13.1585	(0.5923)	12.9344	(1.0902)	16.6230	(0.9027)	0.3250	(0.1157)

* For brevity's sake, the obsolete designations B₁, B₂ are used for thiamin, riboflavin, respectively.

† Slope significantly different from zero ($p < 0.01$).

* Significant difference in slopes between restriction and control group ($p < 0.01$).

42% of the subjects reached deficient α -EGOT levels. However, 67% of them reached deficient plasma PLP levels (results not shown).

Thiamin

For the linear changes in thiamin variables, significant differences were observed between the restriction and the control groups. Regarding total thiamin concentration, the difference in quadratic changes was significant. A decrease in total thiamin concentration and ETK activity and an increase in α -ETK was observed in the restriction group ($p < 0.01$) whereas no changes were observed in the control group (Table 5). A positive quadratic component for total thiamin concentration was found in the restriction group ($p < 0.01$), indicating stabilization after an initial decrease.

Riboflavin

For the linear changes in riboflavin variables, significant differences were observed between the restriction and control groups. Regarding EGR activity and α -EGR, the differences in quadratic changes between restriction and control groups were significant. A decrease in FAD concentration and EGR activity and an increase in α -EGR were observed in the restriction group ($p < 0.01$). Increases in FAD concentration and EGR activity were observed in the control group ($p < 0.01$) (Table 5). A positive quadratic component for EGR activity was observed in the restriction group ($p < 0.01$), indicating stabilization after an initial decrease. A negative quadratic component for α -EGR was found in the restriction group ($p < 0.01$), indicating stabilization after an initial increase.

Vitamin B₆

For the linear changes in vitamin B₆ variables, differences were observed between the restriction and control groups ($p < 0.01$). Regarding PLP concentration, the difference in quadratic changes between the restriction and control groups was significant. A decrease in PLP concentration and EGOT activity and an increase in α -EGOT was observed in the restriction group ($p < 0.01$). A positive quadratic component for PLP concentration was observed in the restriction group, indicating stabilization after an initial decrease. An increase in PLP concentration was observed in the control group ($p < 0.01$) (Table 5).

Urinary Vitamin Excretion

At the start of the experimental period none of the subjects showed urinary excretions of thiamin, riboflavin and 4-pyridoxic acid at levels indicative of vitamin deficiencies. However, during the period of low vitamin intake all restricted subjects reached deficient urinary vitamin levels [31].

The linear and quadratic changes in excretion of thiamin, riboflavin, and 4-pyridoxic acid differed significantly between the restriction and control groups. The excretion of thiamin, riboflavin and 4-pyridoxic acid showed a significant decrease in the restriction group (Table 6). A positive quadratic component for thiamin, riboflavin, and 4-pyridoxic acid was observed in the restriction group ($p < 0.01$), indicating stabilization of urinary excretion after an initial decrease.

Physical Performance

For aerobic power, OBLA, anaerobic power and related variables, no significant vitamin-specific main effects were

Table 7. Maximal Performance: Mean Values (SEM) and Average Slopes (SEM), all Groups Taken Together (n = 24)

Variable†	Week of Study			slope
	0	5	11	
Maximal workload (W)	316 (6.0)	310 (6.2)	296 (6.8)	-1.82 (0.5102)*
$\dot{V}O_{2\text{-max}}$ (l/min)	3.88 (0.07)	3.69 (0.07)	3.42 (0.07)	-0.0417 (0.0052)*
$\dot{V}O_{2\text{-max}}$ (ml · min ⁻¹ · kg ⁻¹)	52.2 (1.28)	51.1 (1.20)	48.7 (1.21)	-0.333 (0.0762)*
Heart rate-max (beats/min)	196 (1.5)	196 (1.7)	194 (1.5)	-0.145 (0.0693)
RER-max	1.08 (0.010)	1.11 (0.008)	1.11 (0.007)	0.0033 (0.001)*
$\dot{V}E/\dot{V}O_{2\text{-max}}$	33.4 (0.84)	34.0 (0.80)	35.7 (0.85)	0.209 (0.0632)*

* Slope significantly different from zero ($p < 0.01$).

† $\dot{V}O_{2\text{-max}}$, aerobic power; RER, respiratory exchange ratio; $\dot{V}E/\dot{V}O_{2\text{-max}}$, respiratory equivalent.

observed (Tables 7, 8 and 9). Therefore, the pooled values are presented.

Maximal Performance

The pooled linear changes in maximal workload, aerobic power ($\dot{V}O_{2\text{-max}}$ and $\dot{V}O_{2\text{-max}}$ /kg body weight), respiratory exchange ratio (RER) and ventilatory equivalent ($\dot{V}E/\dot{V}O_2$) differed significantly from zero. The pooled linear changes in maximal heart rate failed to reach the level of significance chosen in this study ($p < 0.05$) (Table 7).

Onset of Blood Lactate Accumulation (OBLA)

The pooled linear changes in OBLA, oxygen consumption at the OBLA level ($\dot{V}O_{2\text{-OBLA}}$ and $\dot{V}O_{2\text{-OBLA}}$ /kg body weight), respiratory gas exchange ratio (RER-OBLA), and ventilatory equivalent ($\dot{V}E/\dot{V}O_{2\text{-OBLA}}$) differed significantly from zero (Table 8).

A pooled increase was observed in RER-OBLA and $\dot{V}E/\dot{V}O_{2\text{-OBLA}}$. The pooled linear changes in heart rate at the OBLA level failed to reach the level of significance applied in this study ($p < 0.05$). Oxygen consumption at the OBLA level, expressed as a percentage of aerobic power ($\dot{V}O_{2\text{-max}}$), and the rating of perceived exertion at OBLA level (RPE) did not change in the experimental period.

Anaerobic Power

The pooled linear decreases in peak power, absolute and per kg body weight, and mean power differed significantly from zero (Table 9). The pooled linear changes in mean power per kg body weight, power decrease and power decrease per kg body weight did not differ significantly from zero.

Adjustment for potentially confounding changes in body weight, hemoglobin concentration, and ambient temperature using analysis of covariance did not affect the

physical performance results. In addition, analysis of physical performance decrements according to the number of vitamins restricted did not show any significant differences among the four groups. The power of this analysis to detect effects is quite low due to the small number of observations, especially in the non-vitamin- and 3-vitamin-restricted groups (n = 3). Nevertheless, the slopes of aerobic power and oxygen consumption at OBLA level differed significantly from zero in the three single or multiple vitamin-deficient groups.

DISCUSSION

As has been shown in a previous experiment [3], a state of depletion of thiamin, riboflavin and vitamin B₆ can be induced by a diet composed of common food products within 11 weeks. This situation was apparent from decreased vitamin concentrations in blood, decreased erythrocytic enzyme activities, elevation of activation coefficients of these enzymes, and lower urinary vitamin excretion. However, no vitamin-specific clinical signs and symptoms of deficiency were observed. The levels of a number of vitamin variables in blood and urine appeared to stabilize after an initial period of decrease or increase, suggesting that a steady state had been reached in the period of low vitamin intake. Although the vitamin levels in blood and urine were comparable with our previous findings [3], the magnitudes of decrease or increase in blood were smaller, especially for riboflavin and vitamin B₆. This observation is in line with the higher intakes of riboflavin and vitamin B₆ in the present experiment. However, this is contradicted by the stronger decrease in urinary excretion of riboflavin and 4-pyridoxic acid.

Thiamin and/or riboflavin intake, either expressed as mg/day or as mg/MJ, was comparable with that of other depletion studies [32–40]. The vitamin B₆ content of the

Table 8. OBLA and Related Variables: mean values (SEM) and average slopes (SEM), all Groups Taken Together (n = 24)

Variable†	Week Of Study			slope
	0	5	11	
OBLA (W)	240.1 (5.98)	236.0 (7.62)	222.4 (6.93)	-1.63 (0.4649)*
$\dot{V}O_2$ -OBLA (l/min)	3.12 (0.077)	2.95 (0.086)	2.74 (0.079)	-0.0349 (0.0056)*
$\dot{V}O_2$ -OBLA (ml · min ⁻¹ · kg ⁻¹)	42.0 (1.29)	40.9 (1.35)	38.9 (1.19)	-1.288 (0.0843)*
$\dot{V}O_2$ -OBLA/ $\dot{V}O_2$ -max (%)	80.3 (1.42)	79.8 (1.41)	79.9 (1.38)	-0.04 (0.1503)
$\dot{V}E/\dot{V}O_2$ -OBLA	26.1 (0.74)	27.0 (0.54)	28.0 (0.50)	0.172 (0.0534)*
RER-OBLA	0.98 (0.008)	1.02 (0.007)	1.02 (0.007)	0.0035 (0.001)*
Heart rate-OBLA (beats/min)	179.1 (2.03)	177.8 (2.05)	174.8 (1.52)	0.40 (0.1753)
RPE-OBLA	14.9 (0.33)	14.9 (0.37)	14.6 (0.34)	-0.047 (0.0280)

* Slope significantly different from zero (p < 0.01).

† OBLA, onset of blood lactate accumulation; $\dot{V}O_2$ -OBLA, oxygen consumption at OBLA level; $\dot{V}O_2$ -OBLA/ $\dot{V}O_2$ -max, oxygen consumption at OBLA level expressed as a percentage of aerobic power; $\dot{V}E/\dot{V}O_2$ -OBLA, respiratory equivalent at OBLA level; RER-OBLA, respiratory exchange ratio at OBLA level; RPE-OBLA, rate of perceived exertion at OBLA level.

present diet was considerably lower than in one of these other studies [37]. The decrease in urinary excretion of thiamin, riboflavin and/or 4-pyridoxic acid is supporting evidence of adherence to the diet by all subjects. The excretion figures in the period of low vitamin intake were comparable or lower than those in our previous experiment [3]. Moreover, the urinary excretion data were similar to, or lower than, results of other depletion studies in man [32–37,40]; the same holds for ETK and α -ETK values in blood [40].

Although much effort was put into adjustment of energy intake as required to control body weight, losses of weight, percentage of body fat and fat-free mass were observed in

almost all subjects. Loss of body fat was responsible for the greater part of the decrease of body weight. These losses were comparable to similar diet-controlled studies [3,6,41,42].

As iron intake was more than adequate, 10.9 mg dietary iron + 10.0 mg iron from the mineral and trace element supplement, and not more than 150 mL of blood per person was drawn during the experiment, no plausible explanation is available for the decrease in Hb concentration and Ht reading in all subjects. What is more, the ferritin concentration remained unaffected.

Despite a pooled decrease in performance, no vitamin-specific main effects were observed in aerobic power, OBLA, anaerobic power and related variables. Even after

Table 9. Anaerobic Power Output Values: Mean Values (SEM) and Average Slopes (SEM), all Groups Taken Together (n = 24)

Variable	Week of Study			slope
	0	5	11	
Peak power (W)	780.5 (13.47)	762.5 (12.74)	705.6 (15.3)	-6.90 (1.364)*
Peak power (W · kg ⁻¹)	10.5 (0.15)	10.6 (0.16)	10.0 (0.21)	-0.042 (0.0174)*
Mean power (W)	632.7 (10.07)	618.8 (9.61)	587.6 (12.18)	-4.1 (1.17)*
Mean power (W · kg ⁻¹)	8.5 (0.10)	8.6 (0.13)	8.3 (0.16)	-0.014 (0.0147)
Power decrease (W · s ⁻¹)	16.3 (0.95)	14.4 (0.66)	14.4 (0.67)	-0.164 (0.0937)
Power decrease (W · s ⁻¹ · kg ⁻¹)	0.22 (0.01)	0.20 (0.01)	0.20 (0.01)	-0.0011 (0.0012)

* Slope significantly different from zero (p < 0.01).

adjustment for confounding factors such as changes in body weight, Hb concentration and ambient temperature, which may affect physical performance [43–45], no such main effects were found.

The mean decrease in aerobic power was comparable with our previous experiment, while the mean decreases in OBLA and related variables were lower [3]. The range of interindividual differences in response was comparable for aerobic power variables. For OBLA and related variables, the range of interindividual response differences was smaller in the present experiment. Again, no significant correlation was found between the magnitude of decrease of physical performance parameters and biochemical indicators of vitamin status. As the subjects did not change their daily activity patterns, it is unlikely that the variability in functional decrements can be attributed to changes in physical activity, or that potentially detrimental functional effects of marginal vitamin deficiency have been counteracted partly by increased levels of physical activity.

Another interesting finding is that the OBLA-related rating of perceived exertion (RPE) remained constant and was practically comparable to a previous observation [6]. Although an incremental bicycle exercise test was applied in one study [40], most physical performance tests used in the present experiment have not been applied in other vitamin-depletion studies. Therefore, the physical performance decrements in the present experiment are hard to compare with literature data. It has to be noted that our observations relate only to measurements taken during relatively short-term exercise. Measurements of long-term aerobic endurance capacity might have given different results under the conditions of our study.

We questioned why the observed decrements in physical performance could not be ascribed to marginal deficiency in one of the vitamins studied. The absence of main effects, additive effects and interactions and the presence of a pooled decrease in physical performance suggest that marginal deficiency in one or more vitamins studied induces functional decrements of the same order of magnitude. This hypothesis is based on two assumptions. First, the induced metabolic deregulations and the resulting functional decrements are the same for each of the vitamins. Second, the induced degree of marginal deficiency in each vitamin is the rate-limiting factor regarding the metabolic and functional consequences. As the B-vitamins studied are involved in mitochondrial energy production, we hypothesize that the performance decrements may be attributed to decreased mitochondrial metabolism. Because biochemical confirmation of this hypothesis is scarce, the following interpretation may be speculative.

Thiamin, riboflavin, and vitamin B₆ play versatile roles in mitochondrial β -oxidation, carbohydrate and amino acid metabolism, and in the respiratory chain [4,5]. Single deficiencies, especially of thiamin and riboflavin, may

decrease the citric acid cycle function and decrease the activities of the respiratory chain [5,46–48]. Little information is available on metabolic disorders in single marginal deficiency [49,50], and no information at all on the metabolic consequences of combined marginal deficiency. Thiamin deficiency results in depression of the activities of the pyruvate and α -ketoglutarate dehydrogenase complexes [48]. Metabolic consequences include an increase in pyruvate and lactate and a decreased activity of the citric acid cycle [48]. Besides a defective β -oxidation, riboflavin deficiency may result in decreased activities of components of mitochondrial electron transport [46], and diminished carnitine status [51]. Regarding the diminished carnitine status, it is not clear whether the synthesis is impaired or it is secondary to impaired fatty acid oxidation [51,52]. As amino acids do not contribute substantially to energy supply during exercise [53], it is more likely that performance reduction reflects the supposed involvement of vitamin B₆ in lipid metabolism [54]. The precise involvement of vitamin B₆ has not been clearly identified and may be a secondary effect as there is evidence that vitamin B₆ is required in carnitine synthesis [51,55]. In the regulation of the mitochondrial acetyl-CoA/CoA ratio carnitine normally increases the availability of free CoA, thereby prompting the flux of substrates along the citric acid cycle [56]. Decreased carnitine status increases the acetyl-CoA/CoA ratio [57] which inhibits the pyruvate dehydrogenase complex and results in increased lactate concentrations in body fluids [58]. Therefore, if carnitine status is impaired by riboflavin and vitamin B₆ deficiency, the blood lactate level may increase prematurely and fatigue may occur at an earlier time [59]. As thiamin deficiency also results in early rise of lactate, the mitochondrial dysfunctions induced by thiamin, riboflavin and vitamin B₆ deficiencies may cause a reduced muscle tissue redox state [60]. This common feature may form the metabolic base for a functionally similar reduction of physical performance. In addition, the correspondence between changes in physical performance parameters observed in the present experiment and those observed in genetic mitochondrial enzyme defects [61,62] may support our mitochondrial dysfunction hypothesis.

CONCLUSION

In the present study, restriction of either thiamin, riboflavin or vitamin B₆, as well as interactive effects between these vitamins, could not unequivocally be associated with the observed decrements in single bouts of intensive short-duration exercise. However, the analyses yield a plausible hypothesis for a causal role of marginal B-complex vitamin deficiency in the observed decrements. We hypothesize that the absence of vitamin-specific effects on performance

may be caused by quantitatively similar but non-additive effects of the B-vitamins studied on mitochondrial handling of reducing equivalents and consequently on muscle tissue redox state.

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